Kevin Owens

Lead AI & Full Stack Software Engineer

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SUMMARY

Senior Full-Stack & AI Engineer with 15+ years of experience architecting and shipping production-grade applications, automation tools, and intelligent systems end-to-end. Deep expertise in Python, SQL, REST APIs, and AWS (S3, IAM), paired with hands-on experience building AI agents and agentic workflows using AWS Bedrock, Claude, and LLM orchestration frameworks like LangGraph. Focused on turning technical requirements into working automation, from AI triage tools to process automation that reduces manual overhead and speeds up response times. Strong foundation in algorithms, data structures, and architectural decision-making for building tools that scale, combined with a track record of owning deliverables from design through testing and production support. Comfortable working closely with project managers and subject matter experts to translate real operational pain points into reliable, deployed solutions.

EXPERIENCE

Lead AI & Full-Stack Software Engineer

Protech Staffing | Contract

11/2023 - 07/2026 | Remote

- Led a cross-functional team of **40+ engineers** across **4 concurrent product streams**, owning architecture and full-stack delivery for MyTutor's UK and US tutoring platforms built on **React**, **Next.js**, and **Node.js**
- Directed the migration from a monolithic codebase to **AWS-based microservices**, **serverless Lambda** functions, and event-driven workflows, using **SQS** and **EventBridge** to decouple services and support new market expansion
- Integrated **OpenAI** and **Claude APIs** to power automated tutor-student matching and grouping, designing the prompt orchestration and business-rule layer that turned model output into reliable scheduling decisions
- Built the platform's **REST** and **GraphQL** API layer in **TypeScript** and **Python**, connecting React front-end applications to scheduling, classroom, authentication, and payments services
- Delivered the online lesson experience, including secure **WebRTC** video sessions, shared lesson spaces, whiteboards, and cross-device state synchronization
- Led integrations with **Clever**, **ADP**, **Zoom**, **Google Analytics**, and **AWS Amplify**, establishing secure, well-documented API boundaries across teams
- Set technical direction and mentored the team through **Scrum**-based releases, automated testing, and **CI/CD** pipelines, driving an initial market-ready launch in approximately **two months** for a platform serving **250K+ learners**, **1,500+ schools**, and **4M+ lessons**

Full-Stack AI Software Engineer

Walmart Global Tech | Full-time

03/2019- 09/2023 | Remote

- Led architecture and production delivery of an **Agentic AI** financial reconciliation platform in **Python**, processing high-volume retail transactions, payments, settlements, and exception-management workflows
- Architected a hierarchical Supervisor/Worker multi-agent framework in **LangGraph**, introducing confidence-based self-correction and semantic tool routing that improved tool-selection accuracy from **70% to 95%** while cutting prompt-token consumption by approximately **80%**
- Designed a multi-stage reconciliation engine combining clustering algorithms, deterministic business rules, and **LLM agents**, achieving **90%+ matching accuracy** across **10,000+ daily transactions**
- Built a human-in-the-loop Feedback Agent that captured accept, reject, and correction decisions from reviewers and converted them into validated learning signals, improving recommendations without repeated model retraining
- Developed the supporting full-stack application and AI service layer in **React** and **Python**, integrating agent orchestration, enterprise APIs, transaction data sources, and authentication into a single review workflow
- Established production-grade evaluation, observability, guardrails, and auditability using structured logging and monitoring, providing visibility into accuracy, confidence, latency, and agent decision quality
- Optimized large-scale AI workflows through asynchronous processing, caching, and structured outputs, improving throughput while controlling inference cost across the platform

Software Development Engineer II

Amazon | Part-time

02/2015- 01/2019 | Remote

- Led modernization of business-critical Amazon Payments services, migrating core workloads from **Oracle** to **DynamoDB** and reducing dependence on legacy relational infrastructure
- Redesigned data-access patterns and service boundaries in **Python** and **Java** for cloud-native NoSQL workloads, improving scalability, performance, and operational flexibility across high-volume payment systems
- Refactored distributed services to strengthen modularity, extensibility, testability, and fault isolation across core transaction-processing workflows
- Designed resilient **AWS**-native application components using **Lambda** and **API Gateway**, supporting high-throughput, business-critical payment operations with stronger availability and predictable performance
- Developed incremental migration and coexistence strategies that allowed legacy and modern architectures to run safely side by side during phased production rollout
- Improved **CI/CD** pipelines, automated testing, and deployment workflows, increasing release reliability and reducing risk across production changes
- Contributed to architecture decisions, production troubleshooting, and performance analysis, supporting stable execution of high-volume Amazon Payments services

Full-Stack Software Engineer

Scion Staffing | Full-time

03/2010 - 12/2014 | On-site | Phoenix, AZ

- Contributed to the architecture and development of an enterprise **commerce and payments platform** supporting secure transactions, merchant integrations, loyalty programs, and high-volume retail and hospitality workflows
- Built full-stack application capabilities in **JavaScript** and **Python**, spanning merchant-facing interfaces, backend services, payment workflows, and integration layers connecting **POS systems** to processors
- Developed reusable **REST APIs** and integration components that simplified connectivity across payment services, merchant systems, and third-party commerce applications
- Designed and supported scalable transaction-processing workflows with a strong focus on availability, data integrity, security, and predictable performance under high load
- Helped evolve platform capabilities across in-store, online, and mobile commerce experiences as enterprise customer requirements expanded
- Improved production reliability through stronger error handling, structured logging, automated testing, and operational support for business-critical payment services
- Collaborated across engineering, product, and integration teams to translate complex merchant requirements into production-ready software capabilities

EDUCATION

Master's Degree, Computer Science - Arizona State University | 2007 - 2009

Bachelor's Degree, Computer Science - Arizona State University | 2003 – 2007

TECH STACK

Languages: Python • SQL • TypeScript • JavaScript • Java • PHP

Cloud & DevOps: AWS (S3, IAM) • GitHub • Kubernetes • Docker • CI/CD • Serverless • Event-Driven Architecture • Observability • Automated Testing

AI & GenAI: Agentic AI • AWS Bedrock • Claude • LLMs • RAG • LangGraph • Multi-Agent Systems • OpenAI • Gemini • Prompt Engineering • Human-in-the-Loop AI • Structured Outputs

Backend: FastAPI • REST APIs • Microservices • Async Processing • API Integration • Backend Services

Data & Retrieval: PostgreSQL • DynamoDB • Oracle • SQL/NoSQL • Data Processing • Semantic Retrieval • Embeddings • Vector Search

Frontend: React • TypeScript • JavaScript

Architecture: Distributed Systems • Cloud-Native Architecture • AI Orchestration • Production AI • System Design • Evaluation • Guardrails • Reliability Engineering